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Adversarial Interaction Attacks in Refugee Matching

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Abstract

Recent work on refugee allocation has tried the use of predictive models to estimate integration outcomes and propose matching refugees to locations where they are expected to integrate well. Such approaches can considerably improve allocation efficiency. However, in practice, locations may have incentives to manipulate the outcomes they realise, in order to influence future predictions and resulting allocations. This project investigates strategic behavior in refugee allocation systems using simulations based on real refugee resettlement data from HIAS. We study the emergence of such adversarial interaction attacks and their impact on long-term allocation dynamics. We characterize the resulting equilibria and investigate how anticipating future responses affects strategic behavior. We also explore interventions intended to reduce incentives for manipulation. The results show that strategic behavior can significantly modify allocation outcomes and reduce both efficiency and fairness. Furthermore, anticipating the reactions of other locations speeds up convergence toward inefficient and unfair equilibria. Limiting the responsiveness of the allocation system, on the other hand, reduces incentives for manipulation. Finally, we discuss future directions, including the use of karma-based mechanisms to further discourage strategic behavior.

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Chapter 1

Introduction

The place where a refugee is located has a major and lasting effect on how well they integrate [Bansak et al., 2018]. A person may get a job quickly in one area and remain unemployed for years in another. This can depend on the languages spoken in the local labor market, and the availability of support services. Traditionally, resettlement agencies would assign refugees to locations based on heuristic rules and available capacity. Recent work showed that this can be improved a lot with data [Bansak et al., 2018, Ahani et al., 2021]. They suggest introducing a predictive model that estimates refugee employment outcomes across locations and optimizes the refugee-location matching under capacity constraints [Bansak and Paulson, 2024]. Compared to traditional matching methods, outcome-based refugee matching has been associated with substantial improvements in employment outcomes.

However, these improvements do not account for strategic shifts in interactions on the location side. A predictor never observes a refugee’s true potential; it only sees the outcomes actually realised at each location. Refugees are assigned each round and either find employment or do not. These realised outcomes are then fed back to the model to be retrained for the next round.

Locations have to bear the costs of housing and supporting refugees and are part of the system. Importantly, their future costs will depend on who they are matched with. As such, they have both the ability and the incentive to *manipulate* the system. A location seeking to attract fewer refugees from a given group may provide them with less support, resulting in worse outcomes for these refugees. The predictive model then learns that refugees from this group tend to achieve lower employment rates there and directs future arrivals elsewhere. Following prior work, we refer to this behavior as an *adversarial interaction attack*: the location does not alter the algorithm’s data or code, but deliberately drives it to predict something different by acting on the outcomes that feed the feedback loop Ionescu et al. [2023].

The effectiveness of predictive matching depends on the outcome data reflecting refugees’ true potential. A strategic location does not falsify this data. Instead, it generates the data adversarially, by genuinely providing certain refugees with less support so they are less likely to find work. The recorded outcomes stay accurate, but they now reflect the location’s reduced support rather than the refugees’ true potential, so the model learns to send it fewer such refugees. This makes the assignment less efficient and, above all, gives the affected refugees genuinely worse outcomes. Because a location can target specific groups, this harm falls disproportionately on them.

While the consequences and interventions of strategic behavior in such settings remain poorly understood, we organise the project around three questions:

- **Q1** When is attacking actually profitable for a location?
- **Q2** How does strategic behaviour affect the efficiency and fairness of the matching?

- **Q3** Can we design a mechanism that mitigates or removes the incentive to attack?

In answering these questions, we make the following contributions:

- We formalise the repeated refugee-matching problem as a game between a planner that resolves a matching linear program each round and locations that choose an attack level. (Chapters 3 and 4)
- We give a game-theoretic analysis of the locations' interaction, including pure Nash equilibria and two ways of approximating the locations' forward-looking behaviour: Iterated best response (IBR) and model-predictive control (MPC) (Chapter 5).
- Using simulations calibrated on real HIAS resettlement data, we show that strategic manipulation emerges as the equilibrium behavior and that the incentive to attack is largely independent of the discount factor. We then quantify both the resulting efficiency loss and, more importantly, its uneven impact across refugee groups (Chapters 7 and 8).
- We study two interventions. A rate limit on how fast the planner may change its policy, and a minimum-intake quota. We characterize the range of their tuning parameters in which they actually remove the incentive to interact adversarially (Chapters 7 and 8).

Chapter 2

Related Work

This project is based on a few different lines of research. The most direct is data-driven refugee matching, which is what we extend. We also draw on the literature on strategic behavior against learning systems and on mechanism design, and we borrow tools from agent-based modeling and ideas from algorithmic fairness. We go through each in the following and say how it connects to our work.

2.1 Outcome-based refugee matching

The starting point for this project is the line of work that uses predictive models to improve refugee placement. Bansak et al. [2018] showed that matching refugees to locations based on predicted integration outcomes can substantially improve employment, with reported gains of 40 to 70% over existing assignment practices in the United States and Switzerland. Ahani et al. [2021] combined machine learning with integer optimization in the Annie MOORE system deployed by HIAS, the agency whose data we use here. These methods were later extended to a dynamic, capacity-constrained setting in which refugees arrive over time and the model is retrained as new outcomes appear [Bansak and Paulson, 2024]. Beyond refugee resettlement, data-driven decision-support tools have been applied to other public allocation problems, for instance to help low-income families navigate public school choice [Wilson et al., 2009].

We use the same ingredients: our planner solves the same kind of capacity-constrained matching LP, and our predictor is retrained on observed outcomes between rounds. The one thing we change is the assumption that locations are passive. In all of this work, locations simply receive refugees and their outcomes are taken at face value. We instead let them behave strategically, and ask what this does to the equilibrium, the efficiency, and the fairness of the matching.

2.2 Strategic behaviour against learning systems

Once a decision rule depends on data that interested parties can influence, those parties may act on it. This has been studied from two angles. In *strategic classification* [Hardt et al., 2016] and closely related *evasion attacks* adversarial machine learning evasion attacks [Biggio et al., 2013], an agent tweaks its own features to obtain a more favorable prediction from a fixed classifier. In *performative prediction* [Perdomo et al., 2020], the deployment of a model changes the distribution on which it is later trained, and the question becomes where the resulting retraining loop settles.

Our setting is close to these, but differs in two ways. First, the location does not misreport anything about itself. It does not claim that a placement failed when it actually succeeded; it genuinely provides less support so that the poor outcome actually occurs. The bad data are real, not faked, which is what separates this from an evasion attack. Second, the transient behavior matters here

in a way it usually does not in performative prediction: locations are playing a game against each other, so there is no single converged model the planner can just deploy and forget. We call this setting an *adversarial interaction attack*, as follows Ionescu et al. [2023].

2.3 Mechanism Design and Strategy-Proofness

A natural response to strategic behaviour is to design the *allocation rule* itself so that honesty is optimal. Strategy-proofness is a central concept in matching theory, where the question is whether agents can gain by *misstating* their preferences. The same goal shows up in strategy-proof *resource allocation* without money, where agents are given an artificial currency to keep incentives aligned over many rounds [Gorokh et al., 2021, Elokda et al., 2023]; we come back to this idea as a possible direction in the conclusion.

Our notion of *interaction-proofness* (Chapter 4) is the same idea for our setting: we want an allocation rule under which no location can increase its long-run utility by attacking, whatever the round or the history of play. The difference is that we ask for this in a repeated, learning-in-the-loop setting rather than the usual static one where agents state their preferences once, and we test two concrete interventions against it, a rate limit and a quota floor.

2.4 Agent-Based Modelling and Algorithmic Fairness

Methodologically, our analysis is an instance of agent-based modelling, where system-level outcomes emerge from the simple, self-interested rules that individual agents follow [Bonabeau, 2002]. We use this lens to let the equilibrium attack level and the aggregate employment loss emerge from the locations’ best responses rather than imposing them. Our fairness findings then connect to the algorithmic-fairness literature [Barocas et al., 2023, Dwork et al., 2012]: a recurring concern there is disparate impact, where a rule harms a group without any explicit intent to do so, which is exactly the dynamic, incentive-driven pattern we observe when strategic behaviour concentrates its harm on specific refugee groups.

Chapter 3

Model and Problem Formulation

3.1 System Description

We consider a repeated refugee matching setting in which, at each round t , a government matches a set of refugees to a set of locations. A round t represents a batch of arrivals that are processed together; in practice, a round could be a week, a month, or a quarter. The proportional distribution key (the shares α_ℓ) is enforced over a longer horizon, such as one calendar year, while the allocation LP may be solved more frequently.

Refugees are **non-returning agents**: they enter the system only once and want the best possible integration outcome. Locations are **returning agents**: they participate in every round, bear integration costs, and may act strategically in how they interact with matched refugees. At each round, the government solves a **static assignment problem**.

However, because observed outcomes are used to update future predictions, each assignment and interaction influence subsequent allocations. The true expected integration outcome when refugee i is matched to location ℓ is written $V_{i\ell}^{\text{true}}$. In practice, the government does not know $V_{i\ell}^{\text{true}}$; instead it relies on predicted scores $\hat{V}_{i\ell}^t$ built from historical data and updated over time.

This feedback loop creates incentives for locations to manipulate outcomes (adversarial interaction attacks).

3.2 Notation

We collect the notation here for reference. It follows the convention : i indexes a refugee, ℓ indexes a location, k indexes a refugee *type*, and t indexes the allocation round.

Symbol	Range	Description
<i>Sets and indices</i>		
$\mathcal{R}$	finite set	refugees in the current batch
$\mathcal{L}$	finite set	locations
i	$i \in \mathcal{R}$	index of a refugee
ℓ	$\ell \in \mathcal{L}$	index of a location
k	$k \in \{1, \dots, K\}$	index of a refugee type
t	$t \in \mathbb{N}$	index of an allocation round
<i>Parameters</i>		
r	$r \in \mathbb{N}$	number of refugees in the current batch
α_ℓ	$[0, 1]$	target capacity (allocation) share of location ℓ
$V_{i\ell}^{\text{true}}$	$[0, 1]$	true expected integration outcome of (i, ℓ)
$\hat{V}_{i\ell}^t$	$[0, 1]$	predicted score used by the learned planner at round t
$\theta_{k\ell}$	$[0, 1]$	true employment probability of type k at ℓ
β	$[0, 1)$	discount factor of a location
c	$\mathbb{R}_+$	per-round support cost of an unintegrated refugee
<i>Decision variables</i>		
$x_{i\ell}$	$[0, 1]$	assignment: fraction of refugee i matched to ℓ
ℓ	$[0, 1]$	attack level chosen by location ℓ
<i>Outcomes and quantities</i>		
$Y_{i\ell}^{\text{out}}$	$\{0, 1\}$	realised (observed) binary integration outcome
$u_{i,\ell}$	$\mathbb{R}$	per-individual utility of location ℓ from hosting i
U_ℓ	$\mathbb{R}$	location ℓ 's (expected discounted) utility
<i>Interventionss (Chapters 7 and 8)</i>		
ρ	$\mathbb{R}_+$	rate-limit budget (max policy change per round)
$q_t(k, \ell)$	$[0, 1]$	share of type- k refugees sent to ℓ in round t
$q_\ell^{\min}$	$\mathbb{N}$	minimum-intake quota (floor) of location ℓ
φ	$[0, 1]$	quota tightness

Table 3.1: Notation used throughout the report.

A note on indexing. We write indices as subscripts, e.g. $V_{i\ell}$, $x_{i\ell}$, $Y_{i\ell}^{\text{out}}$, for readability. In the simulation code, the same quantities are stored with bracket indexing ($\mathbf{V}[\mathbf{i}, \mathbf{1}]$, $\mathbf{x}[\mathbf{i}, \mathbf{1}]$, etc.). The score $V_{i\ell}$ is the one the planner *acts on*; depending on the planner this is either the learned prediction $\hat{V}_{i\ell}^t$ or the true outcome $V_{i\ell}^{\text{true}}$ (see Chapter 6). When we speak of *types*, we mean groups of refugees that share the same observable attributes (Section 4.2).

3.3 Static Allocation Problem

At each round the planner matches the current batch of refugees to locations in order to maximise the total predicted integration outcome. For each pair (i, ℓ) , the score $V_{i\ell}$ gives the value of sending i to ℓ . This score is whichever the planner has access to: the learned prediction $\hat{V}_{i\ell}^t$ for the deployed planner, or the true value $V_{i\ell}^{\text{true}}$ for the oracle benchmarks (see Chapter 6), where $V_{i\ell}^{\text{true}} = \theta_{k\ell}$ when refugee i is of type k .

Because the exact number of arrivals is not known far in advance, capacities are expressed as proportional shares rather than fixed slot counts: each location ℓ has a target share α_ℓ with

$\sum_{\ell} \alpha_{\ell} = 1$, and with r refugees in the batch its capacity for the round is $\lceil \alpha_{\ell} r \rceil$. The planner solves

$$\max_x \quad \sum_{i \in \mathcal{R}} \sum_{\ell \in \mathcal{L}} V_{i\ell} x_{i\ell} \quad (3.1)$$

$$\text{s.t.} \quad \sum_{\ell \in \mathcal{L}} x_{i\ell} = 1, \quad \forall i \in \mathcal{R}, \quad (3.2)$$

$$\sum_{i \in \mathcal{R}} x_{i\ell} \leq \lceil \alpha_{\ell} r \rceil, \quad \forall \ell \in \mathcal{L}, \quad (3.3)$$

$$x_{i\ell} \in [0, 1], \quad \forall i \in \mathcal{R}, \ell \in \mathcal{L}. \quad (3.4)$$

Constraint (3.2) says each refugee is matched to exactly one location, and constraint (3.3) says no location receives more than its proportional share. We solve the linear relaxation and then round each refugee to its highest-weight location; ties are broken at random. This LP defines the static assignment benchmark that is re-solved at every round of the repeated game.

Remark 1 (A Note on Capacities). *The way capacities are enforced varies by country. In Switzerland the government uses a proportional distribution key, so location ℓ receives roughly $\alpha_{\ell} r$ refugees per period. In other countries (e.g. the US) locations have fixed capacity slots. Our formulation covers both: read $\lceil \alpha_{\ell} r \rceil$ as a generic per-round capacity that can be either a hard cap or derived from a proportional key.*

Chapter 4

Location Response and Adversarial Attacks

After the planner solves the allocation problem, each location sees which refugees it received and interacts with them through the integration process. This chapter describes what a location’s decision looks like within a round, how we model its utility, and how a strategic location can attack the learning loop.

4.1 Setup and Timing

At the start of round t , the planner matches the current batch of refugees to locations by solving the LP of Section 3.3. Each location then interacts with its matched refugees. The timing within a single round is:

1. **Matching stage.** The planner observes the predicted scores $\hat{V}_{i\ell}^t$ and solves the matching LP to obtain the assignment $x_{i\ell}$.
2. **Interaction stage.** Each location ℓ sees its matched refugees and chooses how much support to provide. Equivalently, it chooses an attack level λ_ℓ , which we introduce here and define fully in Section 4.5, where we give its range and its interpretation.
3. **Outcome stage.** Integration outcomes $Y_{i\ell}^{\text{out}}$ are realised and recorded, and (for the learned planner) fed back to retrain the predictor to get $\hat{V}_{i\ell}^{t+1}$.

Locations participate in every round (returning agents) and may behave strategically. Refugees play no strategic role once matched.

4.2 Refugee Characteristics

Each refugee is described by observable attributes. In our data refugees fall into $K = 8$ *types* defined by the combination of gender, English proficiency, and work authorisation. The true employment probability depends on the type k and the location ℓ through $\theta_{k\ell}$.

It is useful to split the attributes into two groups:

- **Matching attributes:** features whose effect on integration depends on the location. The main example in Switzerland is language: a French-speaking refugee does much better in a francophone canton than in an agermanophone one.
- **Competitive attributes:** capability features independent of the assigned location, such as age, education level, or work authorisation.

This distinction matters because the attack works only through the matching attributes. A location cannot change a refugee's underlying capability, the part of integration that comes from competitive attributes and travels with the refugee anywhere. What it can affect is how well a given profile does *at that location*, which is exactly what the matching attributes capture. By providing less support to a targeted group, the location lowers the outcomes actually realised for that group, so the predictor learns that the group integrates poorly there and sends fewer such refugees in the future.

4.3 Integration Cost

The planner optimises predicted employment, but the *actual* cost a location bears when hosting a refugee is not fully captured by a single employment number. A location spends resources on language support, housing, and case management. An unemployed refugee keeps generating these costs round after round, whereas an employed one does not.

We assume that supporting an unemployed refugee costs c per round. As a result, the total cost incurred by a location depends on how long the refugee remains unemployed. If a refugee of type k at location ℓ finds employment in each round with probability $\theta_{k\ell}$, the expected discounted number of rounds generating a cost is:

$$\sum_{s \geq 0} \beta^s (1 - \theta_{k\ell})^s = \frac{1}{1 - \beta + \beta \theta_{k\ell}}, \quad (4.1)$$

so the baseline expected discounted cost of hosting them is $c / (1 - \beta + \beta \theta_{k\ell})$. A higher true employment probability means a shorter expected stay and a lower cost. This is the quantity that increases under an attack.

4.4 Location Utility

We take the default model in which a location gets no explicit benefit from hosting and its utility is the negative of the expected discounted cost. Combining the baseline cost of Section 4.3 with the effect of the attack gives the **utility per round** of location ℓ from hosting a refugee i of type k :

$$u_{i,\ell,\lambda} = -c \cdot \frac{1 + \beta \lambda_\ell \theta_{k\ell}}{1 - \beta + \beta \theta_{k\ell}}. \quad (4.2)$$

The denominator comes from the expected discounted stay in (4.1). The additional term in the numerator reflects the impact of the attack.

Without an attack ($\lambda_\ell = 0$), the expression reduces to the usual cost. Increasing λ_ℓ makes utility more negative because refugees stay unemployed for longer and require support for more rounds.

Attacking therefore has an immediate cost. It becomes worthwhile only if it changes future allocations in the attacker's favour (Section 4.5).

The location's utility in a single round is the sum over the refugees it was assigned,

$$U_\ell^t = \sum_{i \in \mathcal{I}_\ell^t} u_{i,\ell,\lambda}, \quad (4.3)$$

where $\mathcal{I}_\ell^t$ is the set of refugees assigned to ℓ at round t .

4.5 Adversarial Interaction Attacks

So far the location's cost depends on the true outcome. But the location can *influence* the outcome it realises, and that is where manipulation enters.

Attack model. Following the adversarial interaction framework of Ionescu et al. [2023], we model a location's strategic choice as an *attack level* $\lambda_\ell \in [0, 1]$ that scales down the realised outcome. A location does not attack every refugee, only the ones it would rather not host. We call a type k *costly* at ℓ if its true employment probability is below the location's median,

$$k \in \mathcal{C}_\ell \iff \theta_{k\ell} < \text{median}(\theta_{1\ell}, \dots, \theta_{K\ell}), \quad (4.4)$$

so that roughly half of the types are costly at each location. The attack degrades only costly types: the observed outcome of refugee i (of type k) at ℓ is drawn as

$$Y_{i\ell}^{\text{out}} \sim \left((1 - \lambda_{i\ell}) V_{i\ell}^{\text{true}} \right), \quad \lambda_{i\ell} = \begin{cases} \lambda_\ell & \text{if } k \in \mathcal{C}_\ell, \\ 0 & \text{otherwise,} \end{cases} \quad (4.5)$$

where $V_{i\ell}^{\text{true}} = \theta_{k\ell}$ for a refugee i of type k . When $\lambda_\ell = 0$ the location cooperates fully and the outcome reflects the true potential. When $\lambda_\ell > 0$ it deliberately makes the outcome worse for its costly types.

The attack targets outcomes, not scores. Locations cannot directly alter the planner's score V . They can only affect the realised outcome Y^{out} . Because these outcomes are later used to retrain the predictor, poorer outcomes lead the model to assign lower scores to the affected refugee-location pairs. As a result, similar refugees are less likely to be assigned to that location in future rounds.

How the attack affects future rounds. The planner never sees V^{true} . It only sees outcomes, which feed the learning loop. A degraded outcome lowers next round's prediction,

$$Y_{i\ell}^{\text{out}} < V_{i\ell}^{\text{true}} \implies \hat{V}_{i\ell}^{t+1} < \hat{V}_{i\ell}^t \implies \text{fewer similar refugees sent to } \ell \text{ at } t+1. \quad (4.6)$$

For a location that loses utility on its costly types, this is exactly the goal: it pays a small extra cost today to be sent fewer costly refugees tomorrow.

Definition 1 (Adversarial interaction attack). *Location ℓ performs an adversarial interaction attack when it chooses $\lambda_\ell > 0$, so that the realised outcome of its costly types satisfies*

$$[Y_{i\ell}^{\text{out}}] = (1 - \lambda_\ell) V_{i\ell}^{\text{true}} < V_{i\ell}^{\text{true}}.$$

Definition 2 (Interaction-proofness). *The allocation system is interaction-proof if no location can increase its expected discounted utility by choosing $\lambda_\ell > 0$, regardless of the round t or the history of play.*

Interaction-proofness is the synonym of strategy-proofness. In mechanism design we want a system in which honest behaviour ($\lambda_\ell = 0$) is optimal not because locations are well-meaning, but because cheating does not pay.

4.6 Expected Discounted Utility

Because locations are returning agents, what they actually care about is not a single round but the discounted sum of their per-round utilities over the horizon. With discount factor β , location ℓ 's objective is

$$U_\ell = \sum_{t=0}^{T-1} \beta^t U_\ell^t = \sum_{t=0}^{T-1} \beta^t \sum_{i \in \mathcal{I}_\ell^t} u_{i,\ell,\lambda}, \quad (4.7)$$

where the utility per round U_ℓ^t is (4.3) and the per individual term is (4.2). This is the term each location maximises when it chooses its attack level. A larger λ_ℓ has two opposite effects. In the short run, it increases the cost of currently hosted refugees by making their integration outcomes worse. In the long run, it can change future assignments, reducing the number of costly refugees sent to the location.

Attacking is profitable when the long-run benefit exceeds the short-run cost. Determining when this happens is one of the main questions addressed in Chapter 7.

Because the resulting dynamic optimization problem is difficult to solve directly, we approximate location behaviour in Chapter 5 using iterated best response and model-predictive control.

4.7 Block Diagram

In Figure 4.1 at each round, the planner solves the matching LP with the score V , refugees are assigned, and outcomes are realised. Two loops then run in parallel. The *learning loop* feeds observed outcomes to the predictor, which updates V for the next round (this loop is active only for the learned planner). The *strategic loop* is the location's: based on the refugees it received, each location estimates its utility U_ℓ and picks the attack level λ_ℓ that maximises it, which it solves with the methods of Chapter 5.

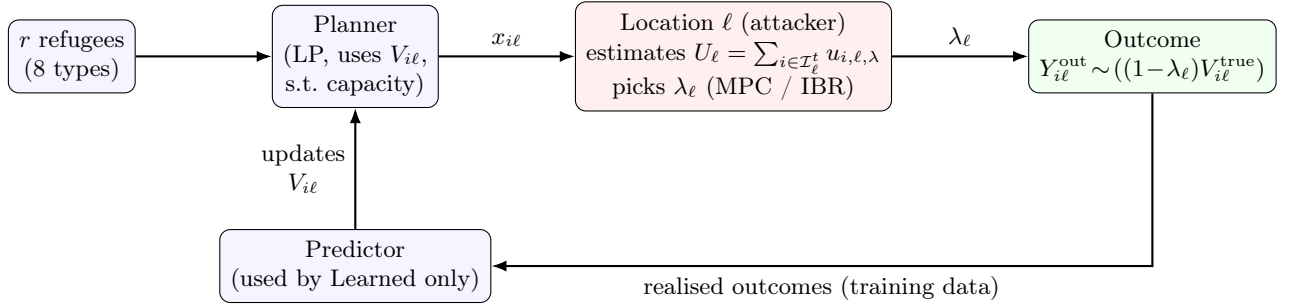


Figure 4.1: Block diagram of strategic outcome-based refugee assignment: the learning loop (outcomes retrain the predictor) and the strategic loop (outcomes feed the location's utility, which sets the next attack level).

4.8 Limitations of the Model

We list the main simplifying assumptions, to keep them in mind when reading the results.

1. **Deterministic vs. stochastic outcomes.** The attack in (4.5) reduces the expected outcome from V^{true} to $(1 - \lambda_\ell)V^{\text{true}}$. While actual outcomes are binary, working with expectations simplifies the analysis and captures the average effect of the attack.
2. **Observation delay.** We assume outcomes are observed at the end of each round and immediately fed to the learner. In practice employment is measured after a delay (e.g. 90 days in Switzerland), which slows the feedback loop but does not change the strategic incentives.
3. **Employment as a measure for integration.** The planner optimises a single outcome (employment), whereas real integration is multi-dimensional. Our model inherits this limitation from the existing algorithmic-matching literature.

Chapter 5

Game-Theoretic Analysis

The locations choose their attack levels to maximise the discounted utility (4.7), and their choices interact: the refugees one location avoids are sent to the others. This makes the interaction a game, and the natural solution is a Nash equilibrium in the attack levels. This chapter defines the equilibrium we look for and the two algorithms we use to find and to interpret it.

5.1 The Attack Game

We restrict attention to a finite set of attack levels,

$$\lambda_\ell \in \Lambda \setminus \{0, 0.25, 0.5, 0.75, 1\}, \quad (5.1)$$

which keeps the game tractable while still spanning “honest” to “full attack”. A pure strategy profile collects one attack level per location, $\lambda = (\lambda_1, \dots, \lambda_{|\mathcal{L}|})$ with each $\lambda_\ell \in \Lambda$. Each location’s payoff is its discounted utility $U_\ell(\lambda)$ from (4.7), which depends on the *whole* profile λ (not just on λ_ℓ).

5.2 Pure Nash Equilibria

Definition 3 (Pure Nash equilibrium). *A profile $\lambda^* \in \Lambda^{|\mathcal{L}|}$ is a pure Nash equilibrium (NE) if no location can improve its discounted utility by unilaterally changing its own attack level:*

$$U_\ell(\lambda_\ell^*, \lambda_{-\ell}^*) \geq U_\ell(\lambda_\ell, \lambda_{-\ell}^*) \quad \forall \lambda_\ell \in \Lambda, \forall \ell \in \mathcal{L}, \quad (5.2)$$

where $\lambda_{-\ell}$ denotes the attack levels of all locations other than ℓ .

In the two-location case the utilities U_{ℓ_0} and U_{ℓ_1} can be laid out as two matrices over the grid $\Lambda \times \Lambda$ (one axis per location’s attack level), and a cell $(\lambda_{\ell_0}, \lambda_{\ell_1})$ is a pure NE exactly when λ_{ℓ_0} is a best response down its column of U_{ℓ_0} and λ_{ℓ_1} is a best response across its row of U_{ℓ_1} . This is how we read the equilibrium off the heatmaps in Chapter 7.

5.3 Iterated Best Response (IBR)

We use iterated best response to model myopic locations that react only to the current behaviour of others. Starting from truthful reporting, each location updates its attack level to its current best response. The process stops when no location wishes to deviate.


```

init       $\lambda \leftarrow (0, \dots, 0)$  // all truthful
repeat
    for each location  $\ell$  in sequence:
         $\lambda_\ell \leftarrow \arg \max_{\lambda'_\ell \in \Lambda} U_\ell(\lambda'_\ell, \lambda_{-\ell})$ 
until  $\lambda$  does not change for a full pass // fixed point

```

Convergence is not guaranteed in general (best-response dynamics can cycle), but on every two-location and sixteen-location HIAS instance we tested, IBR converged to a pure NE. An important diagnostic is the *first* IBR step: it tells us what a location does when everyone else is still truthful, i.e. whether a unilateral attack is already profitable. We use this in Chapter 7 to show that the harm to specific refugee types appears immediately.

5.4 Model-Predictive Control (MPC)

IBR treats a location as reacting to the present. MPC instead models a *forward-looking* attacker that anticipates how the other locations will react *before* committing to its own attack.

General formulation. At each iteration τ , every location $\ell \in \mathcal{L}$ updates its attack level by solving an MPC problem with horizon $H \geq 1$ and discount factor $\beta \in (0, 1]$, anticipating that the opponent $-\ell$ will pick the attack level that maximises its own utility:

$$\begin{aligned}
 \max_{\lambda_\ell^{(0)}, \dots, \lambda_\ell^{(H-1)}} \quad & \sum_{h=0}^{H-1} \beta^h U_\ell(\lambda_\ell^{(h)}, \lambda_{-\ell}^{(h)}) \\
 \text{s.t.} \quad & \lambda_{-\ell}^{(h)} \in \arg \max_{\lambda_{-\ell} \in \Lambda} U_{-\ell}(\lambda_\ell^{(h)}, \lambda_{-\ell}), \quad h = 0, \dots, H-1, \\
 & \lambda_\ell^{(0)} = \lambda_\ell(\tau), \\
 & \lambda_\ell^{(h)} \in \Lambda, \quad h = 0, \dots, H-1,
 \end{aligned} \tag{5.3}$$

where $\Lambda = \{0, 0.25, 0.5, 0.75, 1\}$ is the discrete attack grid, h indexes the steps inside the planning horizon, and $\lambda_{-\ell}$ denotes the attack of the other location(s). The inner $\arg \max$ is the opponent's best response; the outer $\max$ is ℓ 's own choice anticipating that reaction. Location ℓ applies only the first action $\lambda_\ell^{(0)}$ and re-plans at the next iteration. The horizon H controls how far ahead it looks: $H = 1$ recovers the myopic case, and larger H approximates the full discounted objective (4.7).

Specific formulation (used in our simulations). We use $H = 2$ and $\beta = 1$. Because the game is static and the utilities are common knowledge, the MPC problem then reduces to a single bilevel best-response step:

$$\lambda_\ell^{\tau+1} \in \arg \max_{\lambda_\ell \in \Lambda} U_\ell(\lambda_\ell, \lambda_{-\ell}) \quad \text{s.t.} \quad \lambda_{-\ell} \in \arg \max_{\lambda_{-\ell} \in \Lambda} U_{-\ell}(\lambda_\ell, \lambda_{-\ell}). \tag{5.4}$$

For each candidate λ_ℓ , location ℓ anticipates the other location's best response and evaluates its own utility under that reaction, keeping the λ_ℓ that maximises it. In practice this collapses to committing directly to the pure Nash equilibrium of $(U_\ell, U_{-\ell})$ in a single synchronous step Chapter 7.

5.5 Why both

The two algorithms approximate the same discounted objective from opposite ends, and using both makes the conclusions robust. MPC shows that even a far-sighted location, which fully anticipates

the consequences of its attack, still chooses to attack. IBR is much cheaper to compute and converges to a pure NE on every instance we tried, which lets us characterise the equilibrium and trace how it is reached step by step. In the results that follow, the equilibrium attack levels we report agree between the two methods.

Chapter 6

Simulation Setup

All our results come from simulations calibrated on real refugee resettlement data, footnoteThe simulation code is publicly available at <https://github.com/sophialahrech/AIA-refugee-matching>. This chapter describes the data, the planners we compare, and how we aggregate per-round outcomes into the utilities that drive the game.

6.1 HIAS Data

We use resettlement data from HIAS, one of the agencies that has worked on data-driven refugee placement in the United States. From this data we obtain, for each location, the true employment probability $p_{k\ell}$ of each refugee type k , together with each location’s capacity and its realised quarterly arrivals. The data has 16 locations and $K = 8$ refugee types, where a type is the combination of gender, English ability, and work authorisation. The eight types are listed in Table 6.1.

Type k	Sex	English	Work-authorised
0	Female	No	No
1	Male	No	No
2	Female	Yes	No
3	Male	Yes	No
4	Female	No	Yes
5	Male	No	Yes
6	Female	Yes	Yes
7	Male	Yes	Yes

Table 6.1: The eight refugee types, defined by biological sex, English proficiency, and work authorisation.

Calibration. The simulation is calibrated using the HIAS dataset. Employment probabilities $p_{k\ell}$ are estimated from observed outcomes, while refugee arrivals are generated according to the empirical type distribution and quarterly arrival counts. As a result, both the horizon T and the batch sizes r are determined by the data. For each location, the set of costly refugee types $\mathcal{C}_\ell$ is defined using (4.4), reflecting the employment probabilities observed at that location.

6.2 Planners

All three planners solve the same matching problem described in Section 3.3. The only difference between them is the score V used to evaluate refugee location pairs. This allows us to study how different levels of information about outcomes and attacks affect allocation decisions. Each planner serves a distinct purpose and is introduced where it is needed in the analysis.

- **Learned planner** ($= \hat{V}$). A logistic-regression predictor trained on the outcomes observed so far retrained between rounds. *Where.* Used in the multi-location experiments of Chapter 8 and in the evaluation of the quota-floor intervention (Section 7.5). *Why.* This planner relies only on observed outcomes and therefore represents the realistic system that a resettlement agency could deploy in practice. It is also the planner most exposed to the attack, since manipulated outcomes are fed back into the training data and directly affect future predictions.
- **Best-response planner** ($V = (1 - \lambda) V^{\text{true}}$). Knows both the true probabilities V^{true} and the current attacks λ , and so reroutes refugees optimally given the attack. *Where:* the two-location equilibrium and fairness results of Chapter 7. *Why:* it is the strongest possible benchmark : an upper bound on what any planner could do against a known attack. Even with access to V^{true} and λ , the equilibrium remains (1,1). This suggests that the incentive to attack is not driven by imperfect predictions. Because such information is unavailable in practice, this planner serves only as a benchmark.
- **Truthful planner** ($V = (1 - \lambda_{t-1}) V^{\text{true}}$). An observer with oracle access to V^{true} that compensates using the *previous* round's attack as an indicator for the current one. *Where:* the rate-limit intervention in Section 7.4. *Why:* The purpose of the rate-limit intervention is to restrict how quickly the planner can react. The truthful planner already knows V^{true} , so it avoids learning and initialisation effects. It still updates with a one-round delay, making the rate limit ρ the main factor determining the planner's responsiveness.

6.3 Utility Aggregation

For each attack profile λ and seed we run the full horizon and record, at every round t and location ℓ , the per-round utility $U_\ell^t = \sum_{i \in \mathcal{I}_\ell^t} u_{i,\ell,\lambda}$ using the per-individual utility (4.2). We then discount and sum over the horizon to obtain the payoff that defines the game,

$$U_\ell(\lambda) = \sum_{t=0}^{T-1} \beta^t U_\ell^t, \quad (6.1)$$

and average over seeds. Unless stated otherwise we use a per-round support cost $c = 1$ and a discount factor $\beta = 0.9$. For the equilibrium and fairness results we evaluate these utilities under the best-response planner. The intervention results in Sections 7.4 and 7.5 use the planner noted there.

Two outcome metrics. We report two complementary quantities. The *utility* $U_\ell(\lambda)$ is what the locations optimise and what determines the equilibrium (Section 5.2). The *employment rate* is the social-welfare quantity we care about when judging efficiency and fairness. The attack is profitable in the first sense while being harmful in the second, which is the whole tension of the problem.

On the integration cost c . The utility in (4.2) already accounts for the integration cost c , which is fixed to 1 throughout the report. All utility results therefore include integration costs by construction. Employment rates are reported separately as a social-welfare metric.

Chapter 7

Results: Two-Location Setting

We start with two locations playing the attack game against each other. This setting is small enough to plot every cell of the game, which makes the mechanism easy to see. It already shows all the qualitative phenomena we care about. The two locations are a larger and a smaller HIAS location (locations 14 and 4 in our data), and unless stated otherwise we use the best-response planner with $\beta = 0.9$ and $c = 1$.

7.1 The Equilibrium is Mutual Attack

Figure 7.1 shows each location's discounted utility U_ℓ over the full grid of attack levels, with λ_{ℓ_0} on one axis and λ_{ℓ_1} on the other. Reading the game off these two panels (a best response is the best cell *down a column* for ℓ_0 and *across a row* for ℓ_1 , Section 5.2) shows that the unique pure Nash equilibrium is mutual full attack, $\lambda^* = (1, 1)$: the top-right cell, boxed. Each location, acting in its own interest, ends up attacking at the maximum level. The comparison with the truthful outcome is particularly revealing. The profile (0,0) Pareto-dominates the equilibrium: both locations would obtain a higher utility if they remained truthful than under (1,1). The equilibrium therefore exhibits the structure of a social dilemma, where individually optimal decisions lead to a collectively worse outcome. This observation motivates the search for mechanisms that discourage strategic behaviour.

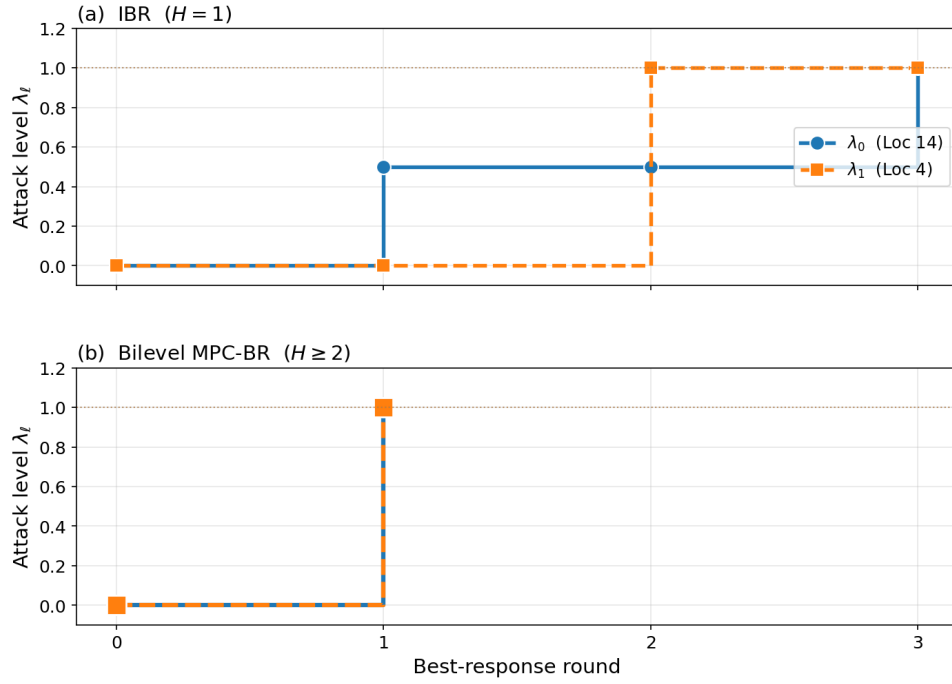


Figure 7.2: Both solution methods reach the same equilibrium. IBR climbs from the truthful corner to $(1, 1)$; the forward-looking MPC best response agrees.

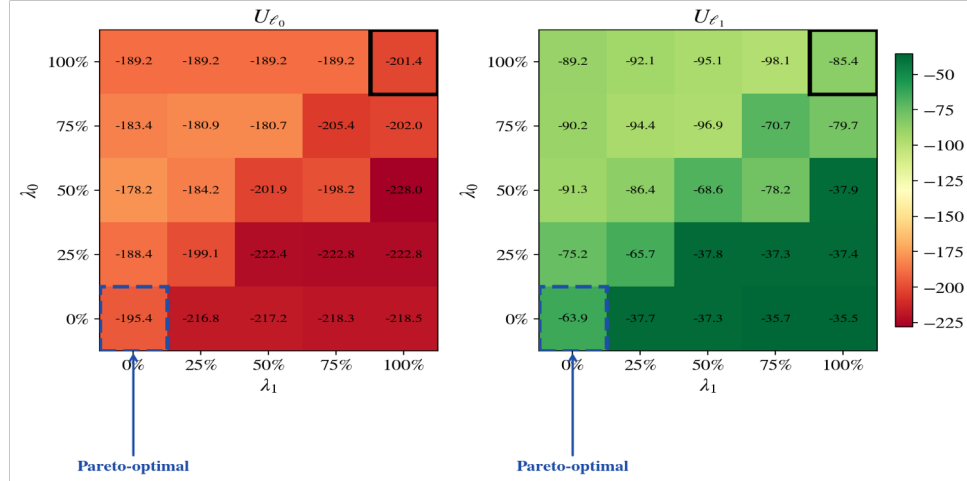


Figure 7.1: Discounted utility of each location over the attack grid (left: U_{ℓ_0} , right: U_{ℓ_1}). The unique pure Nash equilibrium is at $(1, 1)$ (black box). The truthful corner $(0, 0)$ (dashed blue box) is Pareto -optimal.

We confirmed that both equilibrium-finding methods of Chapter 5 agree: iterated best response (Figure 7.2) and the MPC best response finds the same NE equilibrium. So the equilibrium is not an artefact of myopia: a location that fully anticipates the consequences of attacking still attacks.

7.2 Attacking Pays Even for Short-Horizon Locations

Because the benefits of attacking arise in future rounds, one might expect the incentive to attack to depend strongly on the discount factor β . However, Figure 7.3 shows that this is not the case. Across the range of values considered, the equilibrium attack level remains essentially unchanged. Even locations that place relatively little weight on future payoffs still find attacking profitable. The incentive to attack appears robust to the choice of β , suggesting that the result is not driven by unusually forward-looking behaviour.

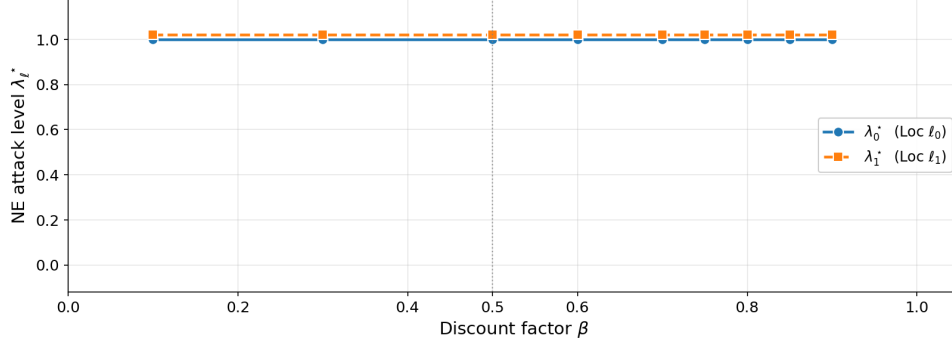


Figure 7.3: The equilibrium attack level barely changes with the discount factor: attacking is profitable even for short-horizon locations.

7.3 Who Gets Hurt: Efficiency and Fairness

We now examine the consequences of strategic behaviour for refugee outcomes. Overall employment decreases from 35% under truthful behaviour to 29% at equilibrium, indicating a loss in aggregate efficiency.

However, this impact is highly uneven across refugee groups. Female, English-speaking, non-work-authorised refugees (type 2) are particularly affected, with employment falling from 39% to 0%. Moreover, much of this decline appears after the very first best-response update, before the system has fully converged to equilibrium.

These results suggest that strategic behaviour can substantially worsen outcomes for specific refugee groups, even when the effect on aggregate employment remains limited.

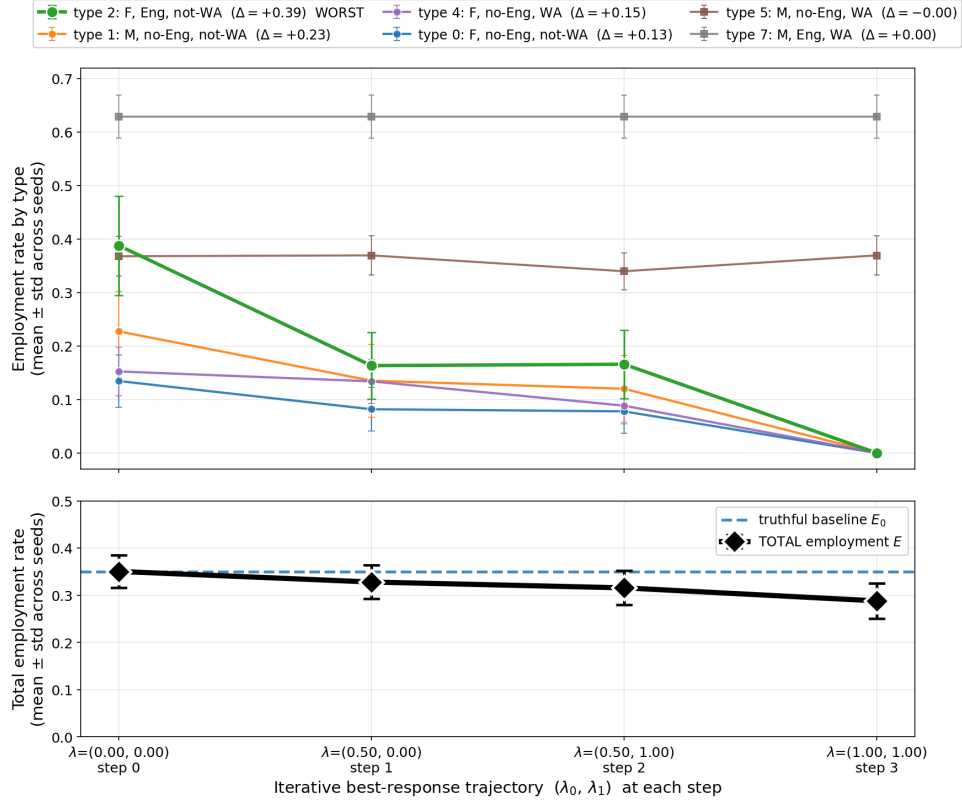


Figure 7.4: Employment rate along the IBR trajectory

7.4 Intervention 1: Rate-Limiting the Planner

The attack works by getting the planner to *react*: a degraded outcome makes the planner reroute the attacked type away. So a first intervention is to stop the planner from reacting too quickly. We bound how much the planner may change the share of each refugee type sent to each location between two consecutive rounds.

Let

$$q_t(k, \ell) = \frac{1}{N_k} \sum_{i: \text{type}_i=k} x_t[i, \ell] \quad (7.1)$$

be the share of type- k refugees sent to ℓ in round t (N_k is the number of type- k refugees that round). The rate-limited planner solves the matching LP (3.1)–(3.4) with the extra constraint

$$|q_t(k, \ell) - q_{t-1}(k, \ell)| \leq \rho \quad \forall k, \ell, \quad (7.2)$$

where the budget ρ controls how fast the policy may move. As $\rho \rightarrow 0$ the policy is frozen and the planner cannot reroute at all. When $\rho \rightarrow \infty$ the constraint is inactive and we recover the unconstrained planner.

The intuition is that if the planner cannot move from one round to the next, the attacker cannot get its costly types rerouted away. Location keeps paying the attack cost for nothing, so attacking stops being profitable. Figure 7.5 confirms this: it sweeps ρ and plots, for each location, the utility of attacking unilaterally against staying truthful. For small enough ρ the attack gain becomes negative, i.e. the system becomes interaction-proof in the sense of Definition 2, around $\rho \approx 0.02$.

The intervention reduces manipulation, but it also slows adaptation to genuine changes. For this reason, ρ should be chosen as a compromise rather than pushed to zero.

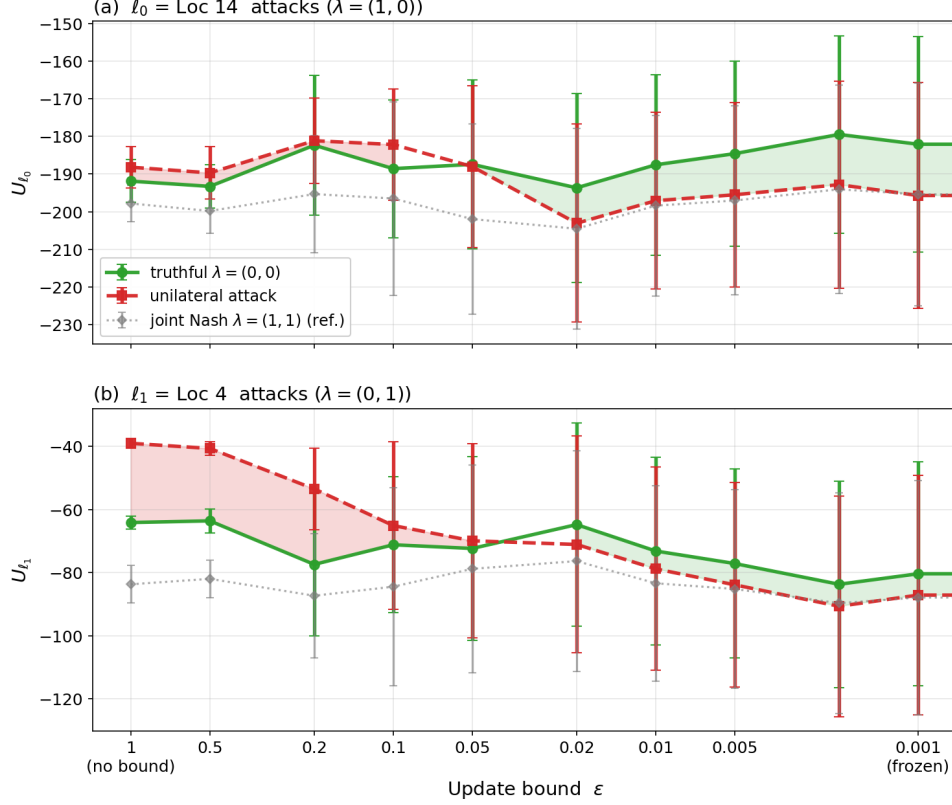


Figure 7.5: intervention 1 (rate limit). Utility of attacking unilaterally vs. staying truthful, as a function of the update budget ρ . As ρ shrinks, attack is not profitable and honesty becomes optimal. We use the truthful planner here to isolate the role of responsiveness.

7.5 Intervention 2: A Minimum-Intake Quota

The second intervention attacks the problem from the allocation side rather than from the planner's speed. We present this intervention here for continuity. It is evaluated on the full 16-location instance with two attackers (locations 9 and 16), since quota constraints become relevant only when redistribution across multiple locations is possible.

The attack works because the planner is free to reroute costly types away. If we force every location to take a guaranteed minimum number of refugees, that freedom shrinks. We give each location a quota floor

$$q_\ell^{\min} = \lfloor \varphi \alpha_\ell r \rfloor, \quad (7.3)$$

a fraction $\varphi \in [0, 1]$ of its expected share, and add it to the matching LP (3.1)–(3.4) as a lower bound:

$$\sum_{i \in \mathcal{R}} x_{i\ell} \geq q_\ell^{\min} \quad \forall \ell \in \mathcal{L}. \quad (7.4)$$

At $\varphi = 1$ the floor equals the capacity, so the assignment is locked on the shares α_ℓ : the scores no longer affect who goes where, attacking is useless. For $\varphi = 0$, the quota floor is inactive and the planner behaves as in the undefended case

The subtle point is that the floor is *per-location*, not *per-type*: it forces a location to take *some* minimum number of refugees, but not of *which* types. So at a light floor like $\varphi = 0.25$ the LP still reroutes the costly types away within the quota. It fills the floor with cheap types and avoids the costly ones. The attacker still gains, and a small floor can even backfire. The intervention only really counts for $\varphi \geq 0.5$, and is total at $\varphi = 1$ (Figure 7.6).

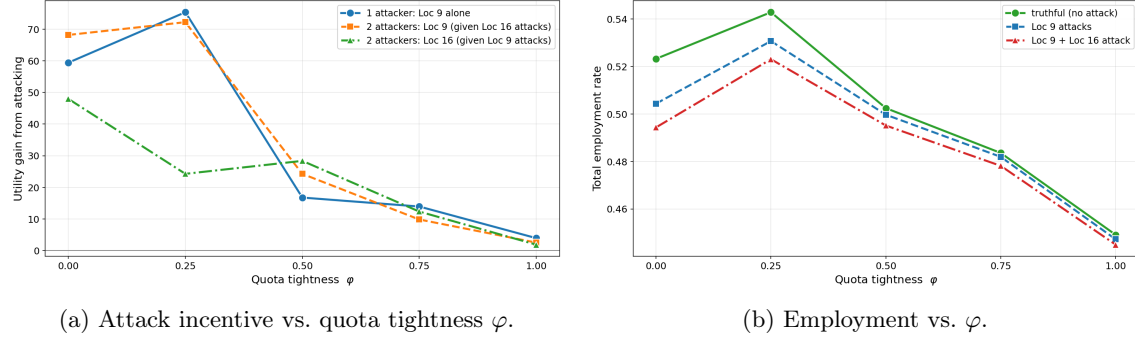


Figure 7.6: intervention 2 (quota floor), evaluated on the 16-location instance with attackers at locations 9 and 16. The intervention is effective for $\varphi \geq 0.5$ and total at $\varphi = 1$.

Chapter 8

Results: Multi-Location Setting

The two-location setting provides a useful illustration of the strategic incentives, but real resettlement systems involve many locations. We therefore turn to the full 16-location HIAS instance to examine whether the same patterns persist at a larger scale. In particular, we consider a setting in which only 2 locations attack while the others remain truthful. As everywhere in the report, the utilities are expressed in the cost-integrated form of Section 6.3 (with $c = 1$).

8.1 Single Attacker

We begin by allowing a single location (location 9) to behave strategically while the other fifteen remain truthful. Figure 8.1 reports its utility and the resulting employment rate as a function of the attack level.

Even in isolation, strategic behaviour increases the location’s utility. Refugees redirected away from location 9 are absorbed by the other locations. The gain is smaller than in the two-location setting because the redirected refugees are distributed across many locations rather than concentrated on a single one.

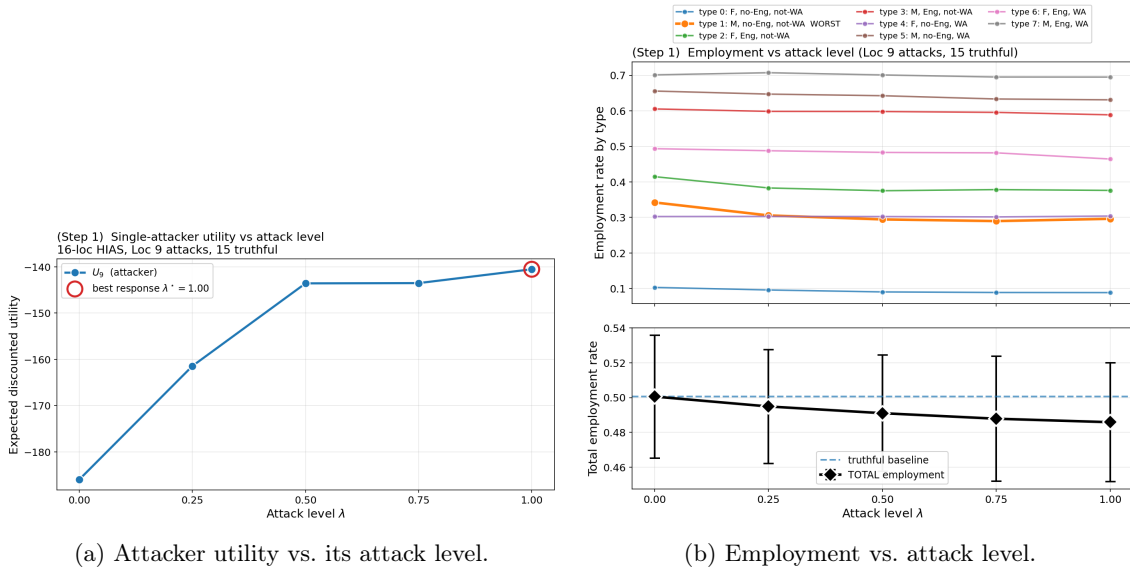


Figure 8.1: Single attacker (location 9) inside the 16-location setting, learned planner. Attacking still raises the attacker’s utility even though the other fifteen locations are truthful.

8.2 Two Attackers

We next consider two strategic locations (locations 9 and 16), while the remaining fourteen locations report truthfully. Figure 8.2 shows the attackers' utilities and the resulting employment levels.

The qualitative behaviour is similar to that observed in the two-location setting. Strategic behaviour remains profitable, and the system converges to a stable equilibrium at $\lambda = (0.75, 0.75)$. However, the equilibrium is less extreme than before, since the costs of rerouting refugees are distributed across fourteen truthful locations.

The impact on employment is also smaller. Aggregate employment decreases from 52% to 49.7%, corresponding to a loss of roughly 2.3 percentage points, compared with about 6 points in the two-location case.

Unlike the two-location setting, the discount factor now plays a role (Figure 8.3). More forward-looking locations choose higher attack levels, suggesting that the benefits of manipulation are realised over a longer horizon when many locations share the rerouted refugees.

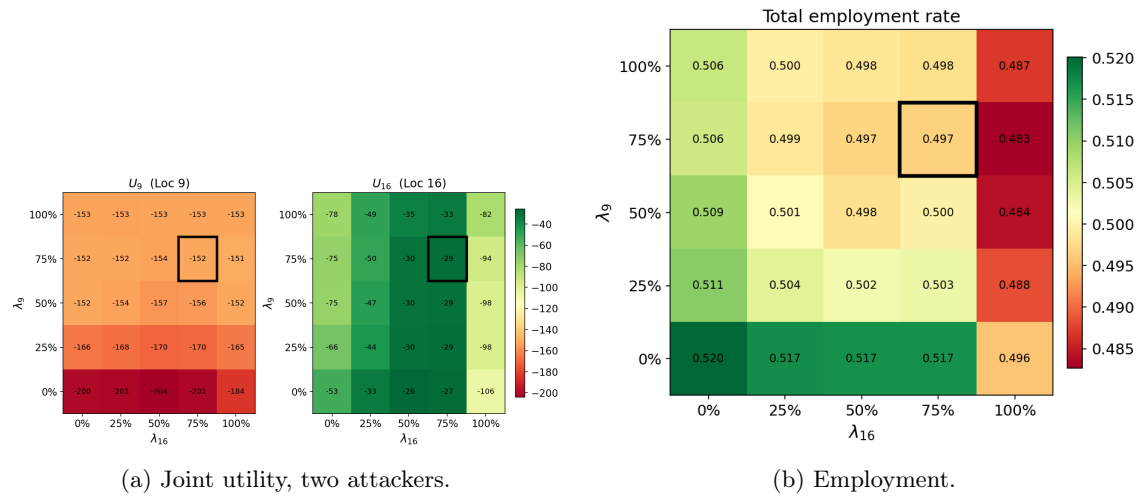


Figure 8.2: Two attackers (locations 9 and 16) in the 16-location setting. The stable equilibrium is $\lambda = (0.75, 0.75)$ and total employment drops from 52% to 49.7%.

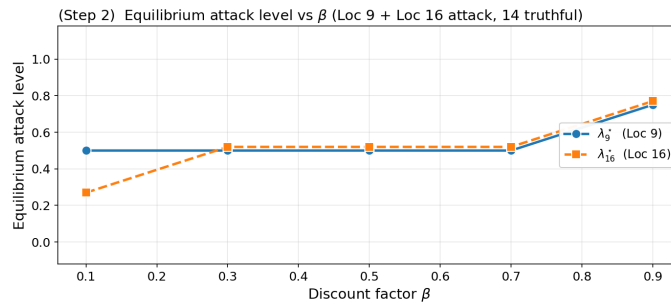


Figure 8.3: Discount-factor sensitivity with many locations: more forward-looking attackers attack harder, in contrast to the two-location case.

8.3 Summary at Scale

The multi-location experiments confirm the main conclusions of the two-location setting. Strategic behaviour remains individually profitable, even when only a single location attacks. However, the equilibrium attack level is lower (0.75 rather than 1), as the costs of rerouting refugees are distributed across many truthful locations.

Aggregate employment declines only moderately, by a few percentage points, but the incentive to manipulate the system persists. The two defence mechanisms reduce this incentive, although both involve a trade-off between robustness and efficiency and require careful tuning of their parameters.

Chapter 9

Conclusion

This project asked whether a learning-based refugee-matching system can be gamed by the locations it serves, what that does to refugees, and whether the mechanism can be fixed. We answer the three questions in turn.

Q1 — When is attacking profitable? Essentially always, at equilibrium. In the two-location game the unique pure Nash equilibrium is mutual full attack, $\lambda^* = (1, 1)$. In the sixteen location setting two attackers settle at $\lambda^* = (0.75, 0.75)$, strong but no longer full because the truthful locations absorb the rerouted load. Alone attacker still gains, and the incentive is essentially independent of how forward-looking the locations are. These results suggest that strategic manipulation is a robust feature of the system rather than an exceptional case.

Q2 — What does it do to efficiency and fairness? Strategic manipulation reduces aggregate employment, although the effect remains moderate (35% $\rightarrow$ 29% in the two-location setting and 52% $\rightarrow$ 49.7% in the 16-location setting with two attackers). This last figure is a lower bound on the damage: when all locations are free to attack, they converge to a near-universal full-attack equilibrium (Chapter A), so the loss would be considerably larger. The impact is also far more pronounced for specific refugee groups. Female, English-speaking, non-work-authorised refugees see their employment rate fall from 39% to 0%, with much of the decline occurring after the first best-response step. Strategic behaviour, therefore, affects both fairness and overall efficiency.

Q3 — Can we remove the incentive? The two interventions reduce the incentive to attack, but both pose trade-offs. Rate limiting makes attacking unprofitable for the smaller location once the update budget falls to approximately $\rho \approx 0.02$. Similarly, a minimum-intake quota becomes effective when the floor reaches at least half of the fair share ($\varphi \geq 0.5$) and fully removes the incentive at $\varphi = 1$.

Both approaches, however, restrict the planner’s flexibility, and this is exactly what costs efficiency. The rate limit slows how fast the planner may change its allocation. This is what stops an attacker from getting its costly types rerouted away. But the same slowness also blocks genuine re-assignments, when job markets or the mix of arriving refugees shift. The quota floor is even more delicate. It forces a minimum intake per location, but not per type. So a small floor ($\varphi = 0.25$) still lets a location shed its costly types while keeping the guaranteed cheap ones. This can leave the targeted refugees worse off than with no floor at all. Only a large floor removes enough of this freedom to help.

Future work. Both interventions we studied work by *constraining the planner*, which is why they are cost-efficient. A natural next direction is to remove the incentive to attack *at its source* instead. For example, one could break the learning loop, so that a location’s future allocation no

longer depends on the outcomes realised there. One promising candidate is a karma mechanism. Each location is given a non-tradable artificial currency. It spends this currency to avoid the refugees it would rather not host, and earns it back by accepting others. If successful, such a mechanism could weaken the link between a location's realised outcomes and its future allocations, so that pushing refugees to a worse outcome than they could have had would no longer buy the location anything.

Future work could examine whether such a mechanism can remove the incentive to attack without reducing efficiency. It would also be interesting to extend the analysis to settings with more strategic locations and to consider integration outcomes beyond employment alone.

Meanwhile, our work suggests that when locations are strategic, focusing on prediction alone is not enough. Refugee matching must be incentive-aware to avoid distributional shifts that could worsen outcomes for all parties involved.

Appendix A

Asynchronous Best Response over All Locations

The equilibria reported in the main text are obtained on a discrete attack grid, either from a two-player payoff matrix (Chapter 7) or with a small number of designated attackers (Chapter 8). A natural question is whether the same picture survives when *all* locations are free to attack and they update their choices one at a time, reacting to each other. This appendix runs that experiment as a robustness check.

A.1 Update Rule

We let all 16 HIAS locations be potential attackers and have them play *asynchronous* iterated best response (an asynchronous, Gauss–Seidel update, as opposed to a synchronous one where every location would move at once). The attack profile $\lambda = (\lambda_1, \dots, \lambda_{16})$ is updated as follows:

```
initialise  $\lambda$     // all truthful, or all full attack
repeat (one sweep)
  for each location  $\ell = 1, \dots, 16$  in turn:
     $\lambda_\ell \leftarrow \arg \max_{\lambda \in \Lambda} U_\ell(\lambda, \lambda_{-\ell})$ 
until a full sweep leaves  $\lambda$  unchanged    // pure Nash equilibrium
```

The update is *asynchronous*: when location ℓ best-responds, the profile $\lambda_{-\ell}$ it reacts to already contains the *updated* attack levels of the locations that moved earlier in the same sweep, not their values from the previous sweep. Each candidate utility $U_\ell(\lambda, \lambda_{-\ell})$ is computed as the discounted cumulative utility defined in Eq. (4.7). The value is estimated by simulating the full planning horizon under the strategy-aware planner and averaging across 12 common-random-number seeds ($\beta = 0.9$, $c = 1$). Unlike the two-location setting, where the complete payoff matrix can be enumerated, the 16-location game contains 5^{16} strategy profiles, making exhaustive tabulation infeasible. Instead, during each best-response update, U_ℓ is evaluated on demand for the five candidate values of λ_ℓ , while holding the strategies of all other locations fixed.

A.2 Result: Convergence to a Common Equilibrium

Figure A.1 shows the trajectory of every location’s attack level across the asynchronous updates, started from two opposite profiles. From the all-truthful start (panel a) the attacks climb sweep by sweep and settle after four sweeps; from the all-attack start (panel b) they settle after two. Both

runs converge to the *same* equilibrium:

$$\lambda_\ell^* = 1 \quad \text{for all locations except location 12,} \quad \lambda_{12}^* = 0.75, \quad (\text{A.1})$$

i.e. all 16 locations attack, 15 of them at the maximum level. The single exception, location 12, is the smallest location in the setting (capacity share $\alpha_{12} \approx 0.001$): with almost no refugees to avoid it has little to gain from the last increment of attack. Its best response saturates just below the maximum.

Asynchronous IBR over all 16 locations converges to the same equilibrium from both starting profiles

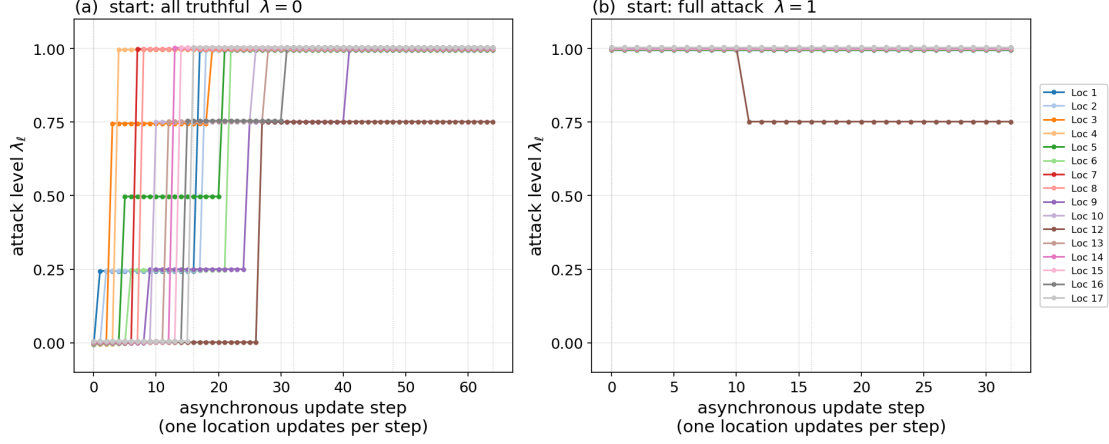


Figure A.1: Asynchronous IBR over all 16 locations. Each curve is one location’s attack level λ_ℓ along the asynchronous updates (one location updates per step; dotted lines mark sweep boundaries). (a) starting from all truthful, (b) starting from full attack.

A.3 Takeaway

Allowing all locations to behave strategically leads to the same qualitative outcome as in Chapters 7 and 8. The system converges to a near-universal full-attack equilibrium, and the same equilibrium is reached from very different initial attack profiles. This suggests that the result is robust to both the starting conditions and the update order.

Compared with the designated-attacker experiments of Chapter 8, the equilibrium is even more extreme. Because there are no truthful locations left to absorb the rerouted refugees, nearly all locations attack at the highest level.

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